Leakage-Aware Structure–Phenotype Prediction of Mechanism-of-Action Profiles Across a Mapped Drug Cohort

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Abstract

Mechanism-of-action (MoA) prediction from molecular structure is often evaluated under random splits that can place chemically identical or closely related compounds on both sides of a test boundary. We constructed an auditable mapping between 3 289 drug-level LISH-MoA records and molecular structures, resolved salts, assigned identical canonical structures to collision groups, and evaluated a graph neural network, two graph–descriptor fusion models, and a pretrained molecular transformer under collision-group-disjoint cross-validation. Under this leakage-controlled protocol, all molecular-arm macro-AUROCs remained close to chance (0.501–0.508) and well below the phenotype-only reference (AUROC 0.636 19). The best molecular-arm log loss was obtained by GIN-TFP (0.023 34), but low log loss accompanied near-chance ranking, indicating calibrated but non-discriminative predictions. These results do not demonstrate that molecular representations are universally inferior; they indicate that, under this mapped cohort, endpoint, and evaluation boundary, structure alone did not provide a convincing predictive advantage for MoA-associated labels. The benchmark addresses prediction of observed MoA-associated annotations and does not establish causal mechanism, target engagement, or biological activity.

1 Introduction

Mechanism-of-action prediction is a multi-label problem in which one compound may be associated with several pharmacological annotations. Molecular structure and cellular phenotype describe different levels of this problem: structure encodes chemical constitution, whereas phenotype summarizes responses observed across cellular systems and experimental conditions [12]. A useful benchmark should therefore distinguish predictive performance from biological mechanism assignment and should evaluate generalization under boundaries that reflect the intended deployment setting.

Random drug-level cross-validation can be optimistic when repeated records, salts, or identical canonical structures cross train and test sets. Collision-aware evaluation is particularly important for multi-label drug data because labels and covariates can be shared across records that are chemically identical [4]. Recent benchmarks indicate that representation scale does not guarantee improved molecular prediction: classical machine learning on fingerprints wins

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47.4 % of task–metric entries in a 156-comparison study [3], and almost no neural embedding statistically improves over ECFP in a 25-embedding comparison [5]. The present study asks whether molecular representations add predictive information to a drug-level MoA-associated phenotype benchmark when canonical-structure collisions are prevented from crossing the primary evaluation boundary. We compare a graph model, two graph–descriptor fusion models, and a pretrained molecular transformer under a common cohort and fold contract. The analysis is a representation benchmark, not a causal MoA study.

2 Materials and methods

2.1 Study design and endpoint

The dataset comprised 3 289 drug-level rows and 206 scored MoA-associated labels from the LISH MoA training release [8]. The primary endpoint was mean column-wise log loss, computed by averaging binary log losses over labels. Secondary endpoints were macro-AUROC [6], macro-AUPRC [2], mean Brier score, and mean expected calibration error (ECE) [9]. AUROC and AUPRC were calculated only for labels with both classes in the test fold; this denominator was retained in the fold records. Probabilities were clipped to $[10^{-7}, 1 - 10^{-7}]$ for log-loss evaluation.

2.2 Structure mapping and collision control

A versioned mapping from `drug_id` to SMILES was constructed from PRISM-aligned records. All 3 289 drug-level rows were covered. Multi-component entries were split, the largest fragment by heavy-atom count was retained, and structures were canonicalized with RDKit [7]. The validated mapping contained 1 722 unique molecules and 794 collision groups. The frozen mapping contract and its hash are recorded in the project data manifest. Collision groups, rather than individual rows, were assigned to folds; no identical canonical structure was allowed to occur in both training and test data.

2.3 Models

Five model configurations were evaluated:

- **Phenotype reference**: an unweighted per-label logistic model fitted independently for each of the 206 labels, using the predefined gene-expression, viability, and experimental-condition features. This model operates on the same drug-level records but uses no molecular structure information.
- **ECFP4–RF**: a random forest on 2 048-bit Morgan radius-2 fingerprints, included as a molecular baseline.
- **GIN**: a graph isomorphism network [13] with 128 hidden units, 3 layers, ReLU activations, batch normalization, dropout $p = 0.1$, mean-pooling readout, and a multi-label output of 206 units.
- **GIN–TFP**: the same GIN with tensor-free persistent descriptors (persistent homology, persistent images, and Betti descriptors; 78 dimensions) concatenated at the head.
- **GIN–TNE**: the same GIN with tensor-network embeddings (192 dimensions) concatenated at the head. Four TNE descriptor failures were retained as zero-vector input rows under an intention-to-treat accounting rule; no failure was removed from the denominator.
- **ChemBERTa**: a pretrained SMILES transformer (`seyonec/ChemBERTa-zinc-base-v1`) [1, 10], fine-tuned with cross-entropy and independent pretrained-weight initialization for every fold.

2.4 Splits and validation

The primary evaluation used five seeds and five collision-group-disjoint folds (25 fold–seed replicates). For each molecular fold, model selection used only training groups: training groups were randomly assigned to six parts and one part was used for validation. Early stopping minimized validation mean column-wise log loss with a patience of 10 epochs. No test group entered model selection. The phenotype baseline was additionally evaluated under the original drug-level k-fold split and a scaffold-held-out sensitivity split [4]. All reported molecular-arm results derive from the same mapped cohort and primary fold definition.

2.5 Training protocol

GIN-family models used a maximum of 50 epochs, batch size 512, learning rate 1×10^{-3}, weight decay 1×10^{-4}, and AdamW optimizer. ChemBERTa used maximum sequence length 128, batch size 32, a maximum of 10 epochs, patience 3, learning rate 2×10^{-5}, and weight decay 0.01. All deep models ran on a single NVIDIA RTX A4000 GPU. For each fold, the pretrained ChemBERTa state was restored before fine-tuning, ensuring no cross-fold weight transfer.

2.6 Reproducibility and analysis boundaries

All mapping records, descriptor computations, fold assignments, and per-fold model outputs were retained with their random seeds, the collision policy, and the descriptor-failure accounting. Three planned analyses were deliberately not performed: per-label calibration diagrams, a bounded quantum-kernel sensitivity analysis, and scaffold-held-out retraining of the four molecular arms. These remain declared extensions rather than negative findings.

3 Results

3.1 Mapping audit

The mapping covered all 3 289 drug-level rows. After salt stripping and canonicalization, all structures passed the RDKit validity gate. The final cohort contained 1 722 unique molecules and 794 collision groups. These collision groups formed the unit of assignment for the primary evaluation.

3.2 Primary collision-group benchmark

Table 1 presents the collision-group results across 25 seed–fold evaluations. GIN-TFP had the lowest molecular-arm log loss at 0.023 34, followed by GIN at 0.024 19 and GIN-TNE at 0.024 34. ChemBERTa had the highest molecular-arm log loss at 0.025 49. Macro-AUROC values for the molecular arms ranged from 0.501 37 to 0.507 85, while the phenotype reference reached 0.636 19. The molecular-arm AUPRC values were also substantially lower than the phenotype reference.

The dissociation between log loss and AUROC is informative: a low multi-label log loss can be achieved by well-calibrated prevalence-like predictions, whereas ranking ability requires discriminating which drugs carry which labels. The structure arms therefore appear reasonably calibrated but non-discriminative on this task. These are descriptive paired-benchmark results; no superiority claim is made without a complete paired statistical comparison.

3.3 Reference and sensitivity analyses

Table 2 compares performance across the three evaluated splits for the phenotype reference and ECFP4–RF baseline. The phenotype reference reproduced the locked random-split result (AUROC 0.643 45) and showed minimal degradation under collision-group splitting (0.636 19)

Table 1: Audited collision-group results across 25 seed–fold evaluations. Log loss is the primary endpoint.

Arm	Log loss	Macro-AUROC	Macro-AUPRC	Brier	ECE
Phenotype reference	0.024 28	0.636 19	0.131 37	0.003 81	0.003 87
ECFP4–RF	0.025 98	0.535 62	0.022 23	0.003 94	0.004 12
GIN	0.024 19	0.507 85	0.013 79	0.003 47	0.004 49
GIN-TFP	0.023 34	0.504 79	0.013 81	0.003 41	0.003 00
GIN-TNE	0.024 34	0.506 01	0.014 60	0.003 41	0.003 06
ChemBERTa	0.025 49	0.501 37	0.013 13	0.003 43	0.007 50

and scaffold sensitivity (0.640 23). The ECFP4–RF baseline reached AUROC 0.535 62 under collision-group splitting and 0.538 17 under scaffold splitting. The phenotype–ECFP4 fusion arm reached AUROC 0.583 23, below the phenotype reference.

Table 2: Split sensitivity for the phenotype reference and ECFP4–RF baseline.

Arm	Random	Collision-group	Scaffold
Phenotype reference	0.643 45	0.636 19	0.640 23
ECFP4–RF	0.535 62	0.535 62	0.538 17

These comparisons reinforce the importance of reporting the split and endpoint together. The modest random-to-collision-group cost for the phenotype reference (0.007 26 AUROC units) reflects the limited impact of collision awareness on this phenotype-driven task, whereas the structure baselines remain near chance across all splits.

3.4 Calibration diagnostics

Post-hoc calibration metrics computed from the archived fold-level test predictions show that ECE roughly doubles from the random split to the scaffold split for every model family, indicating that confidence degrades under scaffold shift even when ranking is only moderately affected. The molecular-arm ECE values (0.003 00–0.007 50) are comparable to the phenotype reference (0.003 87), consistent with the interpretation that molecular predictions are calibrated but non-discriminative. Complete per-configuration calibration metrics are provided in the Supporting Information.

4 Discussion

This benchmark was designed to test a specific question: whether molecular structure representations improve prediction of observed MoA-associated labels under collision-group-disjoint evaluation. The answer is negative for the evaluated molecular arms.

4.1 Calibrated but non-discriminative

GIN-TFP achieved the lowest molecular-arm log loss (0.023 34), marginally below even the phenotype reference on this endpoint, but its macro-AUROC (0.504 79) remained close to chance and far from the phenotype reference (0.636 19). This dissociation is informative: a low multi-label log loss can be achieved by well-calibrated prevalence-like predictions, whereas ranking ability requires discriminating which drugs carry which labels. The structure arms therefore appear reasonably calibrated but non-discriminative on this task.

4.2 Why phenotype features dominate

The phenotype reference performed better on the ranking metrics, which is expected because the labels are associated with cellular response annotations and the phenotype features summarize cellular measurements. This does not make the phenotype model a causal mechanism predictor. Conversely, the structure models do not fail because they are disproven in general; they may require larger chemically diverse cohorts, improved molecular supervision, alternative label definitions, or task-specific pretraining. The present result is therefore an honest benchmark conclusion rather than a universal model ranking.

4.3 Comparison with P5 molecular benchmark

The same model architectures were evaluated on a different task in a companion study of antimalarial activity prediction [11]. Under scaffold-held-out evaluation on 19 836 African antimalarial natural products, ECFP4–RF achieved ROC AUC 0.830 0, exceeding all learned arms. The present P6 benchmark differs in task (MoA labels vs. binary antimalarial activity), label source (phenotype annotations vs. screening activity), and cohort composition (drug-level LISH records vs. natural-product screening data). The consistent ECFP4–RF advantage across both tasks supports the interpretation that local substructure encodings captured by fingerprints remain highly data-efficient for small-to-medium molecular panels, regardless of whether the endpoint is activity classification or MoA-label prediction.

4.4 Implications for representation choice

For comparable drug-level MoA benchmarks, the tested molecular representations did not provide a ranking advantage over the phenotype reference. This result is consistent with broader evidence that classical fingerprint models often remain competitive with graph and sequence models on small-to-medium datasets [3, 5]. The implication is practical: practitioners evaluating molecular representations for MoA-associated label prediction should test against a phenotype baseline under leakage-controlled splits before assuming that more complex representations will improve performance.

The dissociation between log loss and AUROC across all molecular arms is notable. Low log loss with near-chance AUROC indicates that the models produce calibrated probability estimates that approximate the marginal label distribution without learning to discriminate among drugs. This pattern suggests that the molecular features, as currently featurized, do not carry sufficient task-specific signal to override the prevalence-driven baseline for this multi-label phenotype task.

4.5 Collision groups as a contribution

The collision-group mapping is a central contribution of the analysis. It prevents identical canonical structures from appearing in both training and test folds, while retaining duplicated drug identities as an explicit part of the data-generating record. The mapping also exposes a limitation: structure collisions are common, and the effective chemical diversity is closer to 1 722 unique molecules than to the raw row count of 3 289. Four TNE failures were retained as zero-vector intention-to-treat cases, preserving the declared denominator but potentially attenuating descriptor-specific performance.

5 Limitations

First, the benchmark predicts MoA-associated annotations rather than experimentally established causal mechanisms. No target engagement, antimalarial activity, resistance phenotype, or prospective biological validation was measured. Second, the primary molecular comparison used collision-group folds, whereas scaffold-held-out results were not computed for the four new GNN and ChemBERTa arms; the canonical-protocol scaffold sensitivity is available only for the phenotype and ECFP4–RF baselines. Third, summary metrics are reported across paired seed–fold records, but a complete multiplicity-adjusted paired comparison and confidence-interval analysis should accompany any future claim of relative performance. Fourth, per-label calibration diagrams and calibration slope/intercept were not computed; ECE and Brier score are summary calibration diagnostics only. Fifth, the mapping relies on PRISM-aligned source records and a frozen matching procedure; external structure-source replication may alter coverage or collision composition.

6 Future directions

Several extensions could strengthen the evidence base. First, scaffold-held-out retraining of the GNN and ChemBERTa arms would test whether the near-chance AUROCs persist under a stricter generalization boundary; the canonical-protocol scaffold sensitivity is currently available only for the phenotype and ECFP4–RF baselines. Second, per-label calibration analysis would reveal whether specific MoA categories are better predicted by molecular structure than others, potentially identifying a subset of labels where structure contributes. Third, contrastive pretraining on the molecular panel could improve representation quality before fine-tuning. Finally, prospective biochemical validation on a subset of high-confidence predictions would test whether the computational ranking predicts biological activity.

7 Conclusion

Under a frozen, collision-group-disjoint evaluation of 3 289 LISH-MoA drug-level records, the evaluated molecular representations did not demonstrate a predictive advantage over the phenotype reference. GIN-TFP obtained the best molecular-arm log loss, but molecular AUROCs remained close to chance. The result supports reporting collision-aware structure–phenotype benchmarks with explicit failure accounting and cautious interpretation. It does not establish universal molecular-model inferiority or causal biological conclusions.

Data and code availability

All code, frozen splits, per-fold results, and the analysis code supporting the conclusions of this study are available under the MIT license in the public GitHub repository (https://github.com/NanaEngo/Malaria_codesV2). A versioned snapshot will be archived on Zenodo to provide a permanent, citable DOI concurrent with publication.

Author contributions

MVST conceived the study, designed the mapping and benchmark protocol, implemented the GNN and ChemBERTa pipelines, and wrote the manuscript. JPTN contributed to data curation and analysis. SGE supervised the project and contributed to manuscript revision. PS contributed to methodology. WFM contributed to biological context and manuscript revision. All authors reviewed and approved the final version.

Competing interests

The authors declare no competing interests.

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Ethics approval and consent to participate

Not applicable. This study is entirely computational and uses publicly available chemical databases; no human or animal subjects were involved.

Use of Artificial Intelligence

The authors used AI-assisted tools during code development and data-analysis scripting. All affected text and code were reviewed and edited by the authors, who take full responsibility for the article. No AI system contributed as an author.

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